CBP Memo

```
library(tidyverse)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.0     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(lubridate)
library(arrow)
```

```
Warning: package 'arrow' was built under R version 4.4.3
```

```

Attaching package: 'arrow'

The following object is masked from 'package:lubridate':

    duration

The following object is masked from 'package:utils':

    timestamp
```

```r
project_dir <- "/Users/simone/Desktop/DDP/cbp"

encounters_processed_dir <- file.path(
  project_dir, "data/encounters/processed"
)
apprehensions_processed_dir <- file.path(
  project_dir, "data/apprehensions/processed"
)
cross_reference_dir <- file.path(
  project_dir, "inputs/cross-reference-files"
)

apprehensions_latest_path <- file.path(
  apprehensions_processed_dir, "apprehensions-latest.parquet"
)
encounters_latest_path <- file.path(
  encounters_processed_dir, "encounters-latest.parquet"
)
cross_reference_file_path <- file.path(
  cross_reference_dir, "nationwide-encounters-fy20-fy23-aor.csv"
)

apprehensions_latest <- read_parquet(
  apprehensions_latest_path,
  col_select = c(apprehension_datetime, statute_charge))

encounters_latest <- read_parquet(
  encounters_latest_path,
  col_select = c(encounter_datetime, statute_charge))

cross_reference_file <- read_csv(cross_reference_file_path)
```

```
Rows: 53859 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Month Grouping, Month (abbv), Component, Land Border Region, Area ...
dbl  (2): Fiscal Year, Encounter Count

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
encounters_edited <- encounters_latest |>
  mutate(
    title_of_authority = case_when(
      str_detect(statute_charge, "8 USC") ~ "Title 8",
      is.na(statute_charge) ~ "Title 42"
    )
  )

apprehensions_edited <- apprehensions_latest |>
  mutate(
    title_of_authority = case_when(
      str_detect(statute_charge, "8 USC") ~ "Title 8",
      is.na(statute_charge) ~ "Title 42"
    )
  )
```

Intro

United States Border Patrol (USBP) releases apprehension and encounter data in separate sheets. While general encounters include all types of interactions at the border, Title 8 inadmissibles are recorded under the Office of Field Operations (OFO) rather than the USBP. Therefore, the USBP encounters datasets include only Title 8 apprehensions and Title 42 expulsions. On the other hand, USBP apprehensions data covers Title 8 apprehensions only.

Because intended difference between the USBP encounters datasets and the USBP apprehensions datasets is the pandemic-era Title 42 expulsions in the encounters dataset, we aim to consolidate encounters and apprehensions into one, comprehensive dataset that can be filtered depending on `title_of_authority`.

- Original logic:
  - If `statute_charge` contains 8 USC -> Title 8
  - If `statute_charge` is `NA` or missing -> Title 42
  - Otherwise -> NA

Unique Value Exploration

To consolidate these apprehensions and encounters datasets, we attempt to use the `statute_charge` field to indicate whether an encounter is a Title 8 apprehension or Title 42 expulsion. First, we explore the unique values in this field. *To be filled in after law student report?

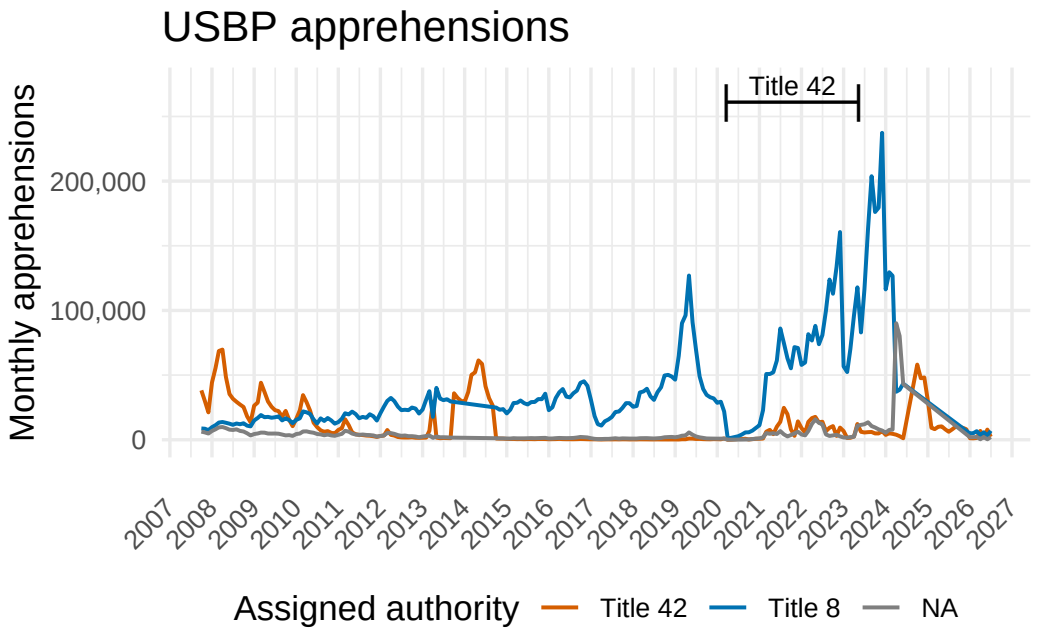
Exploration

USBP Apprehension Counts Over Time

```
monthly_counts <- apprehensions_edited |>
  filter(!is.na(apprehension_datetime)) |>
  mutate(month = floor_date(as.Date(apprehension_datetime), "month")) |>
  count(month, title_of_authority)

bracket_y <- max(monthly_counts$n) * 1.1

ggplot(monthly_counts, aes(month, n, color = title_of_authority)) +
  geom_line(linewidth = 0.7) +
  annotate(
    "segment",
    x = as.Date("2020-03-20"), xend = as.Date("2023-05-11"),
    y = bracket_y, yend = bracket_y
  ) +
  annotate(
    "point",
    x = as.Date(c("2020-03-20", "2023-05-11")),
    y = bracket_y,
    shape = "|", size = 5
  ) +
  annotate(
    "text",
    x = as.Date("2021-10-14"), y = bracket_y * 1.05,
    label = "Title 42", size = 3.5
  ) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(labels = scales::label_comma()) +
  scale_color_manual(
    values = c("Title 8" = "#0072B2", "Title 42" = "#D55E00")
  ) +
  labs(
    x = NULL, y = "Monthly apprehensions", color = "Assigned authority",
    title = "USBP apprehensions"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 0.5))
```



- I applied the same classification rule that labeled rows with 8 USC in `statute_charge` as Title 8, NA `statute_charge` as Title 42, and all else as NA to `apprehensions_latest.parquet`.
- These results lead me to think that `statute_charge` may not be a reliable indication of title of authority because that logic applied here implies that Title 42 exists in apprehensions, which we know not to be true.

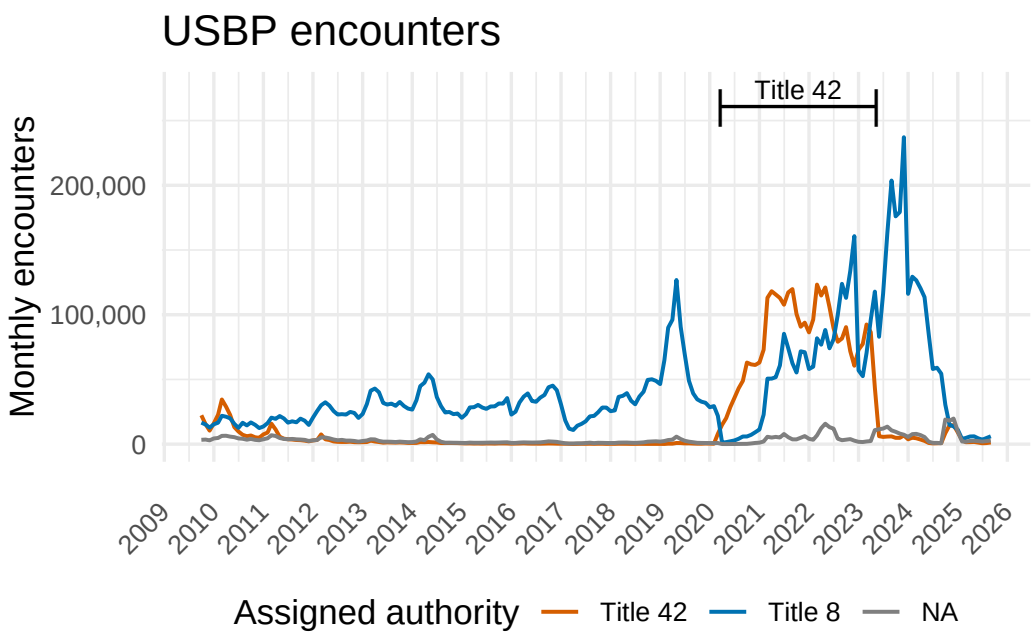
USBP Encounter Apprehension Counts Over Time

```
monthly_counts <- encounters_edited |>
  filter(!is.na(encounter_datetime)) |>
  mutate(month = floor_date(as.Date(encounter_datetime), "month")) |>
  count(month, title_of_authority)

bracket_y <- max(monthly_counts$n) * 1.1

ggplot(monthly_counts, aes(month, n, color = title_of_authority)) +
  geom_line(linewidth = 0.7) +
  annotate(
    "segment",
    x = as.Date("2020-03-20"), xend = as.Date("2023-05-11"),
    y = bracket_y, yend = bracket_y
  ) +
  annotate(
```

```
    "point",
    x = as.Date(c("2020-03-20", "2023-05-11")),
    y = bracket_y,
    shape = "|", size = 5
  ) +
  annotate(
    "text",
    x = as.Date("2021-10-14"), y = bracket_y * 1.05,
    label = "Title 42", size = 3.5
  ) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_continuous(labels = scales::label_comma()) +
  scale_color_manual(
    values = c("Title 8" = "#0072B2", "Title 42" = "#D55E00")
  ) +
  labs(
    x = NULL, y = "Monthly encounters", color = "Assigned authority",
    title = "USBP encounters"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 0.5))
```



```r
# Encounter NA's based on new logic in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11"))) |>
  summarise(sum(is.na(title_of_authority)))
```

```
# A tibble: 1 x 1
  `sum(is.na(title_of_authority))`
                             <int>
1                           155366
```

- The spike during the Title 42 period is promsising.
- There are 154680 NA values during the Title 42 period which could be Title 42 expulsions that we would never know about. How problematic would it be to not provide that info?

Title 8 Encounter and Apprehension Comparison

```r
# Find overlapping date ranges
min(apprehensions_edited$apprehension_datetime)
```

```
[1] "2007-10-01 UTC"
```

```r
max(apprehensions_edited$apprehension_datetime)
```

```
[1] "2026-07-31 23:51:32 UTC"
```

```r
min(encounters_edited$encounter_datetime)
```

```
[1] "2009-10-01 00:01:00 UTC"
```

```r
max(encounters_edited$encounter_datetime)
```

```
[1] "2025-09-30 23:50:00 UTC"
```

```r
# Title 8 encounter count in comparable range
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2009-10-01"),
  as.Date("2025-09-30"))) |>
  summarise(sum(title_of_authority == "Title 8", na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum(title_of_authority == "Title 8", na.rm = TRUE)`
                                                 <int>
1                                              8276838
```

```r
# Apprehension count in comparable range
apprehensions_edited |> filter(between(
  as.Date(apprehension_datetime),
  as.Date("2009-10-01"),
  as.Date("2025-09-30"))) |>
  summarize(n = n())
```

```
# A tibble: 1 x 1
        n
    <int>
1 9540874
```

```r
# Title 42 encounter count in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11")))|>
  summarise(sum((title_of_authority == "Title 42"), na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum((title_of_authority == "Title 42"), na.rm = TRUE)`
                                                   <int>
1                                                3098730
```

```r
# Title 8 encounter count in Title 42 period
encounters_edited |> filter(between(
  as.Date(encounter_datetime),
  as.Date("2020-03-20"),
```

```
  as.Date("2023-05-11"))) |>
  summarise(sum(title_of_authority == "Title 8", na.rm = TRUE))
```

```
# A tibble: 1 x 1
  `sum(title_of_authority == "Title 8", na.rm = TRUE)`
                                                 <int>
1                                              2197266
```

```
# Apprehension count in Title 42 period
apprehensions_edited |> filter(between(
  as.Date(apprehension_datetime),
  as.Date("2020-03-20"),
  as.Date("2023-05-11"))) |>
  summarize(n = n())
```

```
# A tibble: 1 x 1
        n
    <int>
1 2608509
```

- I counted apprehensions and Title 8 encounters within the comparable period (largest period of coverage shared by both datasets) and the Title 42 period.
- In the comparable period, the apprehensions dataset counts 9540874 apprehensions, while the encounters dataset counts 8276838 apprehensions. Assuming the apprehensions dataset is accurate, the logic under-counts apprehensions by 1264036.
- In the Title 42 period, the apprehensions dataset counts 2608509 apprehensions, while the encounters dataset counts only 2197266 apprehensions. Assuming the apprehensions dataset is accurate during the Title 42 period, the logic under-counts apprehensions by 411243.

Title 42 CBP Dashboard Count Comparison

```
start_date <- as.Date("2020-03-20")
end_date <- as.Date("2023-05-11")

official_t42 <- cross_reference_file |>
  filter(
    Component == "U.S. Border Patrol",
    `Title of Authority` == "Title 42"
```

```
  ) |>
  mutate(
    month_number = match(
      str_to_upper(`Month (abbv)`),
      str_to_upper(month.abb)
    ),
    calendar_year = as.integer(`Fiscal Year`) -
      if_else(month_number >= 10L, 1L, 0L),
    month = make_date(calendar_year, month_number, 1)
  ) |>
  filter(between(
    month,
    floor_date(start_date, "month"),
    floor_date(end_date, "month")
  )) |>
  group_by(month) |>
  summarise(
    official_title42 = sum(`Encounter Count`, na.rm = TRUE),
    .groups = "drop"
  )

encounters_t42 <- encounters_edited |>
  filter(between(
    as.Date(encounter_datetime), start_date, end_date),
    title_of_authority == "Title 42"
  ) |>
  mutate(month = floor_date(as.Date(encounter_datetime), "month")) |>
  count(month, name = "assigned_title42")

comparison <- official_t42 |>
  left_join(encounters_t42, by = "month") |>
  mutate(
    difference = assigned_title42 - official_title42,
    pct_difference = 100 * difference / na_if(official_title42, 0)
  )

print(comparison, n = Inf)
```

```
# A tibble: 39 x 5
   month      official_title42 assigned_title42 difference pct_difference
   <date>                <dbl>            <int>      <dbl>          <dbl>
 1 2020-03-01             7094             7181         87           1.23
```

2	2020-04-01	15018	14966	-52	-0.346
3	2020-05-01	20084	20067	-17	-0.0846
4	2020-06-01	28534	28731	197	0.690
5	2020-07-01	35444	35782	338	0.954
6	2020-08-01	42808	43193	385	0.899
7	2020-09-01	48389	48874	485	1.00
8	2020-10-01	63018	63049	31	0.0492
9	2020-11-01	61354	61635	281	0.458
10	2020-12-01	60620	61168	548	0.904
11	2021-01-01	62545	63052	507	0.811
12	2021-02-01	72335	72944	609	0.842
13	2021-03-01	107278	113082	5804	5.41
14	2021-04-01	110869	118157	7288	6.57
15	2021-05-01	111542	115622	4080	3.66
16	2021-06-01	103096	112937	9841	9.55
17	2021-07-01	94264	107733	13469	14.3
18	2021-08-01	93158	117230	24072	25.8
19	2021-09-01	100616	119793	19177	19.1
20	2021-10-01	92741	100604	7863	8.48
21	2021-11-01	87913	90735	2822	3.21
22	2021-12-01	79873	93880	14007	17.5
23	2022-01-01	77541	86354	8813	11.4
24	2022-02-01	90934	96277	5343	5.88
25	2022-03-01	109702	123271	13569	12.4
26	2022-04-01	98362	114836	16474	16.7
27	2022-05-01	103624	121077	17453	16.8
28	2022-06-01	92814	105820	13006	14.0
29	2022-07-01	75800	89485	13685	18.1
30	2022-08-01	72711	79111	6400	8.80
31	2022-09-01	72494	81621	9127	12.6
32	2022-10-01	80097	90414	10317	12.9
33	2022-11-01	69101	71816	2715	3.93
34	2022-12-01	51385	60600	9215	17.9
35	2023-01-01	66538	72951	6413	9.64
36	2023-02-01	76209	77376	1167	1.53
37	2023-03-01	91160	92468	1308	1.43
38	2023-04-01	84203	86496	2293	2.72
39	2023-05-01	31585	38342	6757	21.4

- Percent differences seem negligible at first but generally grow by month throughout the Title 42 period.